Rolling Disk on an Inclined Plane

Classical Mechanics

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1 Introduction

1.1 Overview

This project investigates the motion of a rigid disk on an inclined plane, with emphasis on the distinction between rolling without slipping and slipping motion. Rolling without slipping is defined by the kinematic constraint $v = R\omega$, for which the point of contact between the disk and the plane is momentarily at rest, while slipping occurs when this condition is violated and the contact point slides relative to the surface. Using Newton's laws for translational and rotational dynamics, the equations of motion for both regimes are derived and the role of friction in enforcing rolling motion is examined. The transition from rolling to slipping is identified by comparing the friction required to maintain rolling with the maximum available static friction. Numerical simulations are then used to study the evolution of the disk's linear velocity, angular velocity, and friction force, providing physical insight into the system's behavior [2].

1.2 Problem Statement

A rigid disk moving down an inclined plane may roll without slipping or slip depending on the available friction and the angle of inclination. This project models the disk's motion to determine the conditions governing rolling, slipping, and the transition between these regimes using analytical derivations and numerical simulations.

1.3 Objectives

The objectives of this project are to:

- Define the physical system and assumptions for a rigid disk on an inclined plane.
- Derive the equations of motion for rolling without slipping.
- Derive the equations of motion for slipping motion.
- Determine the transition condition between rolling and slipping.
- Perform numerical simulations to compute the disk's velocity, angular velocity, and friction force.
- Interpret and discuss the results by comparing analytical and numerical findings.

2 Physical System and Assumptions

We consider a rigid disk of mass m and radius R placed on an inclined plane making an angle θ with the horizontal. The disk moves under the influence of gravity and may undergo either rolling without slipping or slipping motion depending on the frictional interaction with the surface. The contact between the disk and the plane is characterized by static and kinetic friction coefficients, denoted by μ_s and μ_k, respectively.

The gravitational force mg acts vertically downward, producing a component $mg\sin\theta$ parallel to the plane and a component $mg\cos\theta$ normal to the plane. The plane exerts a normal reaction force N and a friction force f at the point of contact. The friction force acts as static friction during rolling without slipping and as kinetic friction during slipping.

To simplify the analysis, the following assumptions are made:

- The inclined plane is rigid, fixed, and flat.
- Air resistance and rolling resistance are neglected.

- The motion is planar and the disk does not deform or wobble.
- Friction obeys Coulomb's law with constant coefficients μ_s and μ_k.
- The disk is initially at rest.

3 System Description

A free-body diagram is used to identify all external forces acting on the disk and to relate these forces to the translational acceleration a of the center of mass and the rotational motion about the center. The radius R of the disk plays a crucial role, as it determines the torque generated by friction and links translational and rotational motion during rolling.[4]

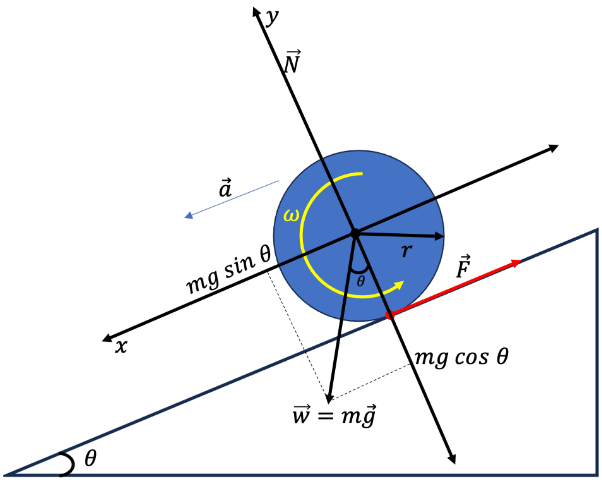

Figure 1: Force Analysis of a Rolling Disk on an Inclined Plane

The forces acting on the disk are:

- **Gravitational force**: The weight mg acts vertically downward through the center of mass. It can be resolved into a component $mg \sin\theta$ parallel to the inclined plane, which drives the motion and contributes to the linear acceleration a, and a component $mg \cos\theta$ perpendicular to the plane.
- **Normal reaction force**: The normal force N acts perpendicular to the surface of the inclined plane. For rigid contact, it balances the normal component of the weight, so that $N = mg \cos\theta$. Since it passes through the center of the disk, it produces no torque.
- **Friction force**: The friction force f acts tangentially at the point of contact and is directed up the plane, opposing relative motion. This force contributes to the net force along the plane and generates a torque of magnitude fr about the center of the disk, where r is the disk radius.

The translational motion of the disk is governed by the net force parallel to the plane,

$$ma = mg \sin\theta - f_s,$$

while the rotational motion is governed by the torque due to friction,

$$I\alpha = f_s r.$$

For rolling without slipping, the linear acceleration a and angular acceleration α are related by the kinematic constraint $a = r\alpha$.

4 Mathematical Model

4.1 Equations of Motion for Pure Rolling

We consider a rigid disk of mass m and radius r rolling down an inclined plane of angle θ without slipping. Let a be the linear acceleration of the disk's center of mass along the plane, ω the angular velocity, and α the angular acceleration. Let f_s denote the (static) friction force acting up the plane[3].

Translational equation (along the plane)

Resolving forces along the incline, the component of gravity down the plane is $mg\sin\theta$ and friction f_s acts up the plane. Hence,

$$ma = mg\sin\theta - f_s. \tag{1}$$

Rotational equation (about the center of mass)

Taking moments about the disk's center, gravity and the normal force pass through the center and produce no torque. The only torque is due to friction:

$$I\alpha = f_s r. \tag{2}$$

For a uniform solid disk,

$$I = \frac{1}{2}mr^2.$$

Rolling constraint (no slip)

For pure rolling, the no-slip condition links translation and rotation:

$$v = r\omega \quad \Rightarrow \quad a = r\alpha.$$

Derived expressions for a, α, and f

Using the rolling constraint $\alpha = \frac{a}{r}$ in the rotational equation gives

$$I\left(\frac{a}{r}\right) = f_s r \quad \Rightarrow \quad f_s = \frac{I}{r^2}\,a.$$

Substituting this into the translational equation yields

$$ma = mg\sin\theta - \frac{I}{r^2}a \quad \Rightarrow \quad a\left(m + \frac{I}{r^2}\right) = mg\sin\theta.$$

Therefore, the linear acceleration is

$$a = \frac{g\sin\theta}{1 + \dfrac{I}{mr^2}}.$$

Using $\alpha = \frac{a}{r}$, the angular acceleration is

$$\alpha = \frac{1}{r}\,\frac{g\sin\theta}{1 + \dfrac{I}{mr^2}}.$$

The friction force is

$$f = \frac{I}{r^2}\,a = \frac{I}{r^2}\,\frac{g\sin\theta}{1 + \dfrac{I}{mr^2}}\,m.$$

Special case: uniform solid disk

For $I = \frac{1}{2}mr^2$, we have $\dfrac{I}{mr^2} = \dfrac{1}{2}$. Substituting gives

$$a = \frac{2}{3}g\sin\theta, \qquad \alpha = \frac{2}{3}\frac{g\sin\theta}{r}, \qquad f_s = \frac{1}{3}mg\sin\theta.$$

In pure rolling, f_s is a static friction force that adjusts to satisfy the no-slip condition; it does not necessarily equal $\mu_s N$, but must satisfy $|f_s| \le \mu_s N$ for rolling to remain valid.

4.2 Slipping (Sliding) Motion

If the available static friction is insufficient to maintain the no-slip condition, the disk begins to slip on the inclined plane. In this regime, the rolling constraint $v = r\omega$ (and hence $a = r\alpha$) no longer applies. Friction is kinetic and has constant magnitude

$$f_k = \mu_k N = \mu_k mg\cos\theta,$$

acting opposite the relative motion at the contact point (typically up the plane if the disk slips downward).

Translational motion (along the plane)

Along the incline, gravity drives the motion and kinetic friction opposes it:

$$ma = mg\sin\theta - f_k.$$

Substituting for f_k gives the linear acceleration

$$a = g\sin\theta - \mu_k g\cos\theta = g\left(\sin\theta - \mu_k\cos\theta\right).$$

Rotational motion (about the center of mass)

About the center of mass, gravity and the normal force pass through the center and produce no torque. The only torque is due to kinetic friction:

$$I\alpha = f_k r.$$

Hence, the angular acceleration is

$$\alpha = \frac{f_k r}{I} = \frac{\mu_k mg\cos\theta\, r}{I}.$$

Special case: uniform solid disk

For a uniform solid disk with $I = \frac{1}{2}mr^2$, the accelerations reduce to

$$a = g\left(\sin\theta - \mu_k\cos\theta\right), \qquad \alpha = \frac{2\mu_k g\cos\theta}{r}.$$

Remark on slip direction

The direction of kinetic friction is determined by the relative motion at the contact point. If the disk slides down the incline, friction acts up the incline. If the relative slip reverses, the friction direction reverses accordingly; however, its magnitude remains $f_k = \mu_k mg\cos\theta$.

4.3 Transition Condition: Rolling → Slipping

Pure rolling is possible only if the contact force can supply the static friction required to enforce the no-slip constraint. Let f_s denote the friction force predicted by the pure rolling dynamics. Rolling breaks down when this required friction exceeds the maximum available static friction.

Maximum available static friction

The maximum static friction is

$$f_k = \mu_s N = \mu_s mg\cos\theta.$$

Required friction for pure rolling

From the pure rolling analysis, the friction force required to maintain rolling is

$$f_s = \frac{I}{r^2}\,a,$$

with rolling acceleration

$$a = \frac{g\sin\theta}{1 + \dfrac{I}{mr^2}}.$$

Substituting gives

$$f_s = \frac{\left(\dfrac{I}{mr^2}\right)}{1 + \left(\dfrac{I}{mr^2}\right)}\,mg\sin\theta.$$

Transition criterion

Pure rolling is maintained when

$$f_s < \mu_s mg\cos\theta$$

and the transition to slipping occurs when

$$f_k = \mu_k mg\cos\theta$$

For a uniform solid disk with $I = \frac{1}{2}mr^2$, the critical condition becomes

$$\mu_s^{\text{crit}} = \frac{1}{3}\tan\theta, \qquad \tan\theta_{\text{crit}} = 3\mu_s.$$

Dynamic transition from slipping to rolling

When the disk is initially slipping, the translational and rotational motions evolve independently under kinetic friction. Slipping ends when the no-slip condition is satisfied,

$$v(t^*) = \omega(t^*)\,r,$$

where t^* denotes the transition time.

Starting from rest with initial angular velocity ω_0, the velocity and angular velocity during slipping are

$$v(t) = (g\sin\theta - \mu_k g\cos\theta)\,t, \qquad \omega(t) = \omega_0 + \frac{2\mu_k g\cos\theta}{r}\,t.$$

The transition time is therefore

$$t^* = \frac{\omega_0 r}{g(\sin\theta - 3\mu_k \cos\theta)}.$$

For $t > t^*$, kinetic friction is replaced by static friction and the disk follows pure rolling motion with acceleration

$$a_{\text{roll}} = \frac{2}{3} g \sin\theta$$

for a uniform solid disk.

5 Energy Analysis

The motion of the disk down the inclined plane is driven by gravity, which converts gravitational potential energy into kinetic energy. The way this energy is distributed depends on whether the disk undergoes pure rolling or slipping motion.

5.1 Energy distribution in pure rolling

For pure rolling motion, the disk satisfies the no-slip condition $v = \omega r$. The total kinetic energy of the disk is the sum of translational and rotational kinetic energy:

$$K_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2.$$

For a uniform solid disk, the moment of inertia is $I = \frac{1}{2}mr^2$. Using $v = \omega r$, the translational kinetic energy is

$$K_{\text{trans}} = \frac{1}{2}mv^2,$$

and the rotational kinetic energy becomes

$$K_{\text{rot}} = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 = \frac{1}{4}mv^2.$$

Hence, the total kinetic energy is

$$K_{\text{total}} = \frac{3}{4}mv^2.$$

The energy distribution in pure rolling is therefore:

- Translational kinetic energy: $\frac{2}{3}$ of the total,
- Rotational kinetic energy: $\frac{1}{3}$ of the total.

Since the point of contact between the disk and the plane is momentarily at rest, static friction does no work. As a result, mechanical energy is conserved during pure rolling, and gravitational potential energy is converted entirely into translational and rotational kinetic energy.

5.2 Energy dissipation during pure slipping

When slipping occurs, the no-slip condition is violated and kinetic friction acts at the contact point. Unlike static friction, kinetic friction does work because the contact point moves relative to the surface, leading to dissipation of mechanical energy in the form of heat.

During slipping, the disk initially possesses only translational kinetic energy,

$$K_{\text{trans}} = \frac{1}{2}mv^2,$$

while the rotational kinetic energy is negligible,

$$K_{\mathrm{rot}} \approx 0.$$

Hence, the total kinetic energy is

$$K_{\mathrm{total}} = \frac{1}{2}mv^2.$$

Including dissipation, the gravitational potential energy lost is converted into kinetic energy and energy dissipated by friction:

$$mgh = K_{\mathrm{total}} + K_{\mathrm{diss}},$$

where K_{diss} represents the energy lost due to kinetic friction. Rearranging,

$$K_{\mathrm{diss}} = \frac{1}{2}mv^2 - mgh.$$

For motion along an inclined plane of angle θ, the vertical height is

$$h = s\sin\theta,$$

where s is the distance traveled along the plane.

Therefore, during slipping, gravitational potential energy is converted into:

- Translational kinetic energy,
- Rotational kinetic energy (as the disk begins to spin),
- Thermal energy due to kinetic friction.

As a result, mechanical energy is not conserved in the slipping regime, which is why the slipping motion is energetically less efficient than pure rolling.

5.3 Energy dissipation during slipping and rolling

Total kinetic energy

The total kinetic energy of the disk is given by the sum of translational and rotational kinetic energy:

$$K_{\mathrm{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2.$$

Using $I = \frac{1}{2}mr^2$ and noting that during slipping $v \neq r\omega$, the kinetic energy components are

$$K_{\mathrm{trans}} = \frac{1}{2}mv^2,$$

$$K_{\mathrm{rot}} = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 = \frac{1}{4}mr^2\omega^2.$$

Hence, the total kinetic energy during slipping is

$$K_{\mathrm{total}} = \frac{1}{2}mv^2 + \frac{1}{4}mr^2\omega^2.$$

Energy loss (friction dissipation)

When slipping occurs, kinetic friction performs work and dissipates energy. The energy lost due to friction is

$$E_{\mathrm{loss}} = f_k\, d_{\mathrm{slip}} = \mu_k mg\cos\theta\, d_{\mathrm{slip}},$$

where d_{slip} is the sliding distance at the contact point.

Key difference

In pure rolling motion, the no-slip condition $v = r\omega$ enforces a fixed ratio between translational and rotational kinetic energy, and static friction does no work. In contrast, during slipping, translational and rotational energies evolve independently and depend on the relative velocity at the contact point, with part of the mechanical energy dissipated as heat.

6 Rolling versus Slipping: Friction, Energy, and Constraints

Property	**Pure Rolling**	**Slipping**
Friction Type	Static f_s	Kinetic f_k
Magnitude	$f_s = \frac{1}{3}mg\sin\theta$ (uniform solid disk)	$f_k = \mu_k mg\cos\theta$
Constraint	$\lvert f_s\rvert < \mu_s mg\cos\theta$ and $v = r\omega$	$f_k = \mu_k mg\cos\theta$ and $v \neq r\omega$
Energy Loss	None (contact point at rest)	Yes (relative motion)
Work by Friction	$W_s = 0$ (no slip $\Rightarrow$ no relative displacement at contact)	$W_k = -f_k\, d_{\text{slip}} = -\mu_k mg\cos\theta\, d_{\text{slip}}$

Table 1: Comparison of pure rolling and slipping motion for a disk on an inclined plane.

7 Numerical Simulation

Numerical simulations were performed to illustrate the time evolution of the disk movement and to highlight the differences between the pure rolling and slipping regimes. The equations of motion derived in the previous sections were numerically integrated over a fixed time interval, starting from rest, using prescribed physical parameters such as the incline angle and friction coefficients [1]. The results of these simulations can be visualized in the video linked here, which demonstrates the rolling and sliding simulation.

7.1 Velocity evolution

Figure 2 shows the center-of-mass velocity $v(t)$ and the tangential velocity $r\omega(t)$ for both pure rolling and slipping motion. In the pure rolling case, the two curves coincide for all times, confirming the no-slip condition

$$v(t) = r\omega(t).$$

This indicates that the point of contact between the disk and the plane is instantaneously at rest.

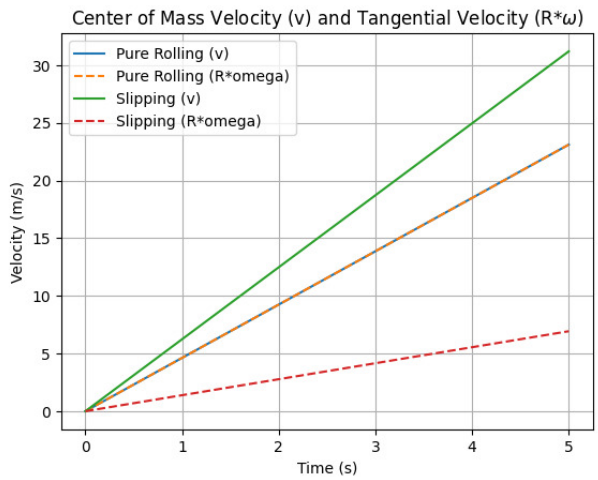

Figure 2: Velocity comparison highlighting the no-slip and slipping regimes.

In contrast, during slipping motion, the center-of-mass velocity $v(t)$ grows faster than $r\omega(t)$. This separation between the two curves indicates relative motion at the contact point and hence violation of the rolling constraint. The larger linear velocity in the slipping case is consistent with the higher translational acceleration predicted analytically.

7.2 Friction force

Figure 3 shows the friction force as a function of time. For pure rolling, the friction force remains constant and lies below the maximum static friction limit $\mu_s N$, indicating that static friction is sufficient to maintain rolling without slipping. Since the contact point is at rest, this friction force does no work.

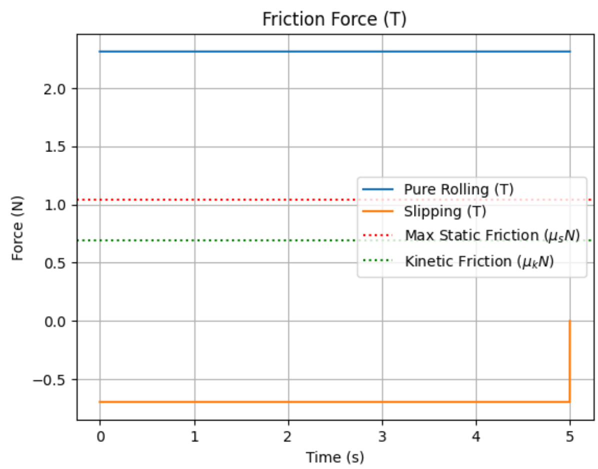

Figure 3: Friction force as a function of time for pure rolling and slipping motion.

For slipping motion, the friction force equals the kinetic friction $f_k = \mu_k N$ and remains constant in magnitude. This force acts opposite the direction of relative motion and results in energy dissipation. The comparison with the static friction threshold clearly demonstrates why rolling cannot be sustained in this regime.

7.3 Position of the center of mass

Figure 4 shows the position of the disk's center of mass as a function of time. The slipping case exhibits a faster increase in position compared to pure rolling, reflecting the larger translational acceleration during slipping. The pure rolling motion shows slower but more efficient acceleration, as part of the gravitational energy is converted into rotational motion.

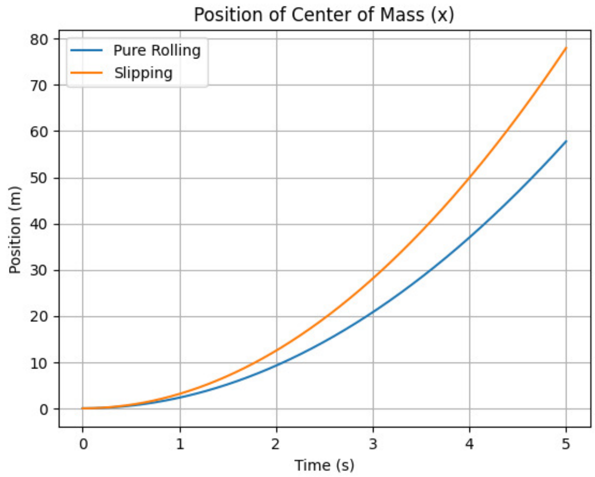

Figure 4: Effect of rolling and slipping on the translational motion of the disk.

7.4 Physical interpretation

These simulations confirm the analytical predictions. Pure rolling is characterized by the kinematic constraint $v = r\omega$, constant static friction, and conservation of mechanical energy. Slipping motion, on the other hand, shows independent evolution of $v(t)$ and $\omega(t)$, constant kinetic friction, and energy dissipation. The numerical results therefore provide clear visual evidence of the fundamental differences between rolling and slipping dynamics.

8 Results

Pure rolling occurs when static friction enforces the no-slip condition $v = r\omega$, giving a constant acceleration $a = \frac{2}{3}g\sin\theta$ and conserving mechanical energy with $\frac{2}{3}$ translational and $\frac{1}{3}$ rotational kinetic energy. When static friction is insufficient, the disk slips, translational and rotational motions decouple, kinetic friction dissipates energy, and acceleration increases but with lower efficiency. The transition between regimes is controlled by the friction limit and the velocity condition $v(t^*) = r\omega(t^*)$, and numerical simulations confirm these analytical results.

Conclusion

The dynamics of a disk on an inclined plane were analyzed for both pure rolling and slipping motion. Pure rolling occurs when static friction is sufficient to enforce the no-slip condition, resulting in conservation of mechanical energy and a fixed distribution between translational and rotational kinetic energy. When this condition is violated, slipping occurs, kinetic friction dissipates energy, and translational and rotational motions evolve independently. The analytical predictions were fully supported by numerical simulations, highlighting the central role of friction in determining the motion and energy behavior of the system.

References

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